

19. Event Link Controller (ELC)

19.1 Overview

The Event Link Controller (ELC) uses the event requests generated by various peripheral modules as source signals to connect them to different modules, allowing direct link between the modules without CPU intervention.

[Table 19.1](#) lists the ELC specifications, and [Figure 19.1](#) shows a block diagram.

Table 19.1 ELC specifications

Item	Description
Event link function	322 types of event signals can be directly connected to modules. The ELC generates the ELC event signal, and events that activate the DTC.
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	Security and Privilege attribution can be set for each registers

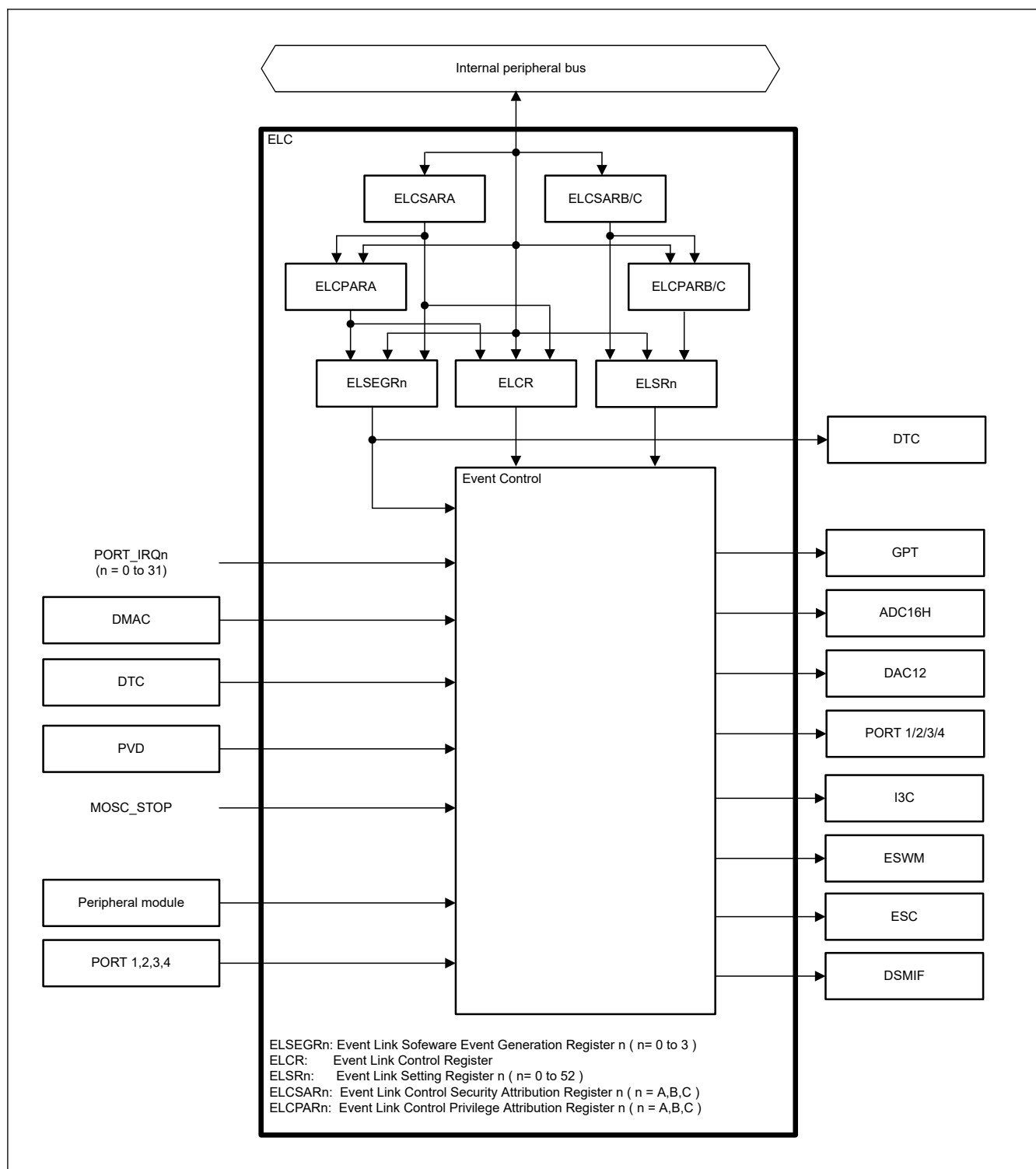


Figure 19.1 ELC block diagram

19.2 Register Descriptions

19.2.1 ELCR : Event Link Controller Register

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x000

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ELCON	—	—	—	—	—	—	—

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.	R/W
7	ELCON	All Event Link Enable 0: ELC function is disabled. 1: ELC function is enabled.	R/W

Note: S-TYPE-3, P-TYPE-3

The ELCR register controls the ELC operation.

19.2.2 ELSEGRn : Event Link Software Event Generation Register n (n = 0 to 3)

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x004 + 0x004 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	WI	WE	—	—	—	—	—	SEG

Value after reset: 1 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	SEG	Software Event Generation 0: Normal operation 1: Software event is generated.	W
5:1	—	These bits are read as 0. The write value should be 0.	R/W
6	WE	SEG Bit Write Enable 0: Write to SEG bit disabled. 1: Write to SEG bit enabled.	R/W
7	WI	ELSEGR Register Write Disable 0: Write to ELSEGR register enabled. 1: Write to ELSEGR register disabled.	W

Note: S-TYPE-3, P-TYPE-3

SEG bit (Software Event Generation)

When 1 is written to the SEG bit while the WE bit is 1, a software event is generated. This bit is read as 0. Even when 1 is written to this bit, data is not stored. The WE bit must be set to 1 before writing to this bit.

A software event can trigger a linked DTC event.

WE bit (SEG Bit Write Enable)

The SEG bit can only be written to when the WE bit is 1. Clear the WI bit to 0 before writing to this bit.

[Setting condition]

- If 1 is written to this bit while the WI bit is 0, this bit becomes 1.

[Clearing condition]

- If 0 is written to this bit while the WI bit is 0, this bit becomes 0.

WI bit (ELSEGR Register Write Disable)

The ELSEGR register can only be written to when the write value to the WI bit is 0. This bit is read as 1. Before setting the WE or SEG bit, the WI bit must be set to 0.

19.2.3 ELSRn:Event Link Setting Register n (n = 0 to 52)

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x020 + 0x004 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	ELS[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	ELS[9:0]	Event Link Select 0x000: Event output disabled for the associated peripheral module 0x001: Number setting for the event signal to be linked ⋮ 0x1F8: Number setting for the event signal to be linked Others: Settings prohibited	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: P-TYPE-3

Note: This register has different secure access permission depending on the setting of Trusted Event Route Control Register (TEVTRCR).
If the security attribution is configured as Secure,

- Secure access is allowed.
- Non-secure write access is ignored and Non-secure read access is read as 0, TrustZone access error is generated.

If the security attribution is configured as Non-secure and the trusted event route is disabled,

- Secure write access is ignored and Secure read access is read as 0, TrustZone access error is generated.
- Non-secure access is allowed.

If the security attribution is configured as Non-secure and the trusted event route is enabled,

- Secure access is allowed.
- Non-secure write access is ignored and Non-secure read access is allowed.
- TrustZone access error is not generated.

The ELSRn register specifies an event signal to be linked to each peripheral module. Table 19.2 shows the association between the ELSRn register and the peripheral modules. Table 19.3 shows the association between the event signal names set in the ELSRn register and the signal numbers.

Table 19.2 Association between the ELSRn registers and peripheral functions (1 of 2)

Register name	Peripheral function (module)	Event name
ELSR0	GPT (A)	ELC_GPTA
ELSR1	GPT (B)	ELC_GPTB
ELSR2	GPT (C)	ELC_GPTC
ELSR3	GPT (D)	ELC_GPTD
ELSR4	GPT (E)	ELC_GPTE
ELSR5	GPT (F)	ELC_GPTF
ELSR6	GPT (G)	ELC_GPTG
ELSR7	GPT (H)	ELC_GPTH

Table 19.2 Association between the ELSRn registers and peripheral functions (2 of 2)

Register name	Peripheral function (module)	Event name
ELSR12	DAC12 channel 0	ELC_DA0
ELSR13	DAC12 channel 1	ELC_DA1
ELSR14	PORT1	ELC_PORT1
ELSR15	PORT2	ELC_PORT2
ELSR16	PORT3	ELC_PORT3
ELSR17	PORT4	ELC_PORT4
ELSR19	ADC16HA0	ELC_HAD00
ELSR20	ADC16HB0	ELC_HAD01
ELSR21	ADC16HC0	ELC_HAD02
ELSR22	ADC16HA1	ELC_HAD10
ELSR23	ADC16HB1	ELC_HAD11
ELSR24	ADC16HC1	ELC_HAD12
ELSR25	ADC16HA2	ELC_HAD20
ELSR26	ADC16HB2	ELC_HAD21
ELSR27	ADC16HC2	ELC_HAD22
ELSR30	I3C	ELC_I3C
ELSR31	ESWM	ELC_GPTP0
ELSR32	ESWM	ELC_GPTP1
ELSR33	ESC	ELC_ESC0
ELSR34	ESC	ELC_ESC1
ELSR35	DSMIF0	ELC_DSMIF0_CAP0
ELSR36	DSMIF0	ELC_DSMIF0_CAP1
ELSR37	DSMIF0	ELC_DSMIF0_CAP2
ELSR38	DSMIF0	ELC_DSMIF0_CAP3
ELSR39	DSMIF0	ELC_DSMIF0_CAP4
ELSR40	DSMIF0	ELC_DSMIF0_CAP5
ELSR41	DSMIF0	ELC_DSMIF0_CNT0
ELSR42	DSMIF0	ELC_DSMIF0_CNT1
ELSR43	DSMIF0	ELC_DSMIF0_CNT3
ELSR44	DSMIF1	ELC_DSMIF1_CAP0
ELSR45	DSMIF1	ELC_DSMIF1_CAP1
ELSR46	DSMIF1	ELC_DSMIF1_CAP2
ELSR47	DSMIF1	ELC_DSMIF1_CAP3
ELSR48	DSMIF1	ELC_DSMIF1_CAP4
ELSR49	DSMIF1	ELC_DSMIF1_CAP5
ELSR50	DSMIF1	ELC_DSMIF1_CNT0
ELSR51	DSMIF1	ELC_DSMIF1_CNT1
ELSR52	DSMIF1	ELC_DSMIF1_CNT3

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (1 of 9)

Event number	Interrupt request source	Name	Description
0x001	Port	PORT_IRQ0 ^{*1}	External pin interrupt 0
0x002		PORT_IRQ1 ^{*1}	External pin interrupt 1
0x003		PORT_IRQ2 ^{*1}	External pin interrupt 2
0x004		PORT_IRQ3 ^{*1}	External pin interrupt 3
0x005		PORT_IRQ4 ^{*1}	External pin interrupt 4
0x006		PORT_IRQ5 ^{*1}	External pin interrupt 5
0x007		PORT_IRQ6 ^{*1}	External pin interrupt 6
0x008		PORT_IRQ7 ^{*1}	External pin interrupt 7
0x009		PORT_IRQ8 ^{*1}	External pin interrupt 8
0x00A		PORT_IRQ9 ^{*1}	External pin interrupt 9
0x00B		PORT_IRQ10 ^{*1}	External pin interrupt 10
0x00C		PORT_IRQ11 ^{*1}	External pin interrupt 11
0x00D		PORT_IRQ12 ^{*1}	External pin interrupt 12
0x00E		PORT_IRQ13 ^{*1}	External pin interrupt 13
0x00F		PORT_IRQ14 ^{*1}	External pin interrupt 14
0x010		PORT_IRQ15 ^{*1}	External pin interrupt 15
0x011		PORT_IRQ16 ^{*1}	External pin interrupt 16
0x012		PORT_IRQ17 ^{*1}	External pin interrupt 17
0x013		PORT_IRQ18 ^{*1}	External pin interrupt 18
0x014		PORT_IRQ19 ^{*1}	External pin interrupt 19
0x015		PORT_IRQ20 ^{*1}	External pin interrupt 20
0x016		PORT_IRQ21 ^{*1}	External pin interrupt 21
0x017		PORT_IRQ22 ^{*1}	External pin interrupt 22
0x018		PORT_IRQ23 ^{*1}	External pin interrupt 23
0x019		PORT_IRQ24 ^{*1}	External pin interrupt 24
0x01A		PORT_IRQ25 ^{*1}	External pin interrupt 25
0x01B		PORT_IRQ26 ^{*1}	External pin interrupt 26
0x01C		PORT_IRQ27 ^{*1}	External pin interrupt 27
0x01D		PORT_IRQ28 ^{*1}	External pin interrupt 28
0x01E		PORT_IRQ29 ^{*1}	External pin interrupt 29
0x01F		PORT_IRQ30 ^{*1}	External pin interrupt 30
0x020		PORT_IRQ31 ^{*1}	External pin interrupt 31

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (2 of 9)

Event number	Interrupt request source	Name	Description
0x040	DMAC0	DMAC00_INT	DMAC0 transfer end 0
0x041		DMAC01_INT	DMAC0 transfer end 1
0x042		DMAC02_INT	DMAC0 transfer end 2
0x043		DMAC03_INT	DMAC0 transfer end 3
0x044		DMAC04_INT	DMAC0 transfer end 4
0x045		DMAC05_INT	DMAC0 transfer end 5
0x046		DMAC06_INT	DMAC0 transfer end 6
0x047		DMAC07_INT	DMAC0 transfer end 7
0x048	DMAC1	DMAC10_INT	DMAC1 transfer end 0
0x049		DMAC11_INT	DMAC1 transfer end 1
0x04A		DMAC12_INT	DMAC1 transfer end 2
0x04B		DMAC13_INT	DMAC1 transfer end 3
0x04C		DMAC14_INT	DMAC1 transfer end 4
0x04D		DMAC15_INT	DMAC1 transfer end 5
0x04E		DMAC16_INT	DMAC1 transfer end 6
0x04F		DMAC17_INT	DMAC1 transfer end 7
0x052	DTC0	DTC0_DTCEND	DTC0 transfer end
0x055	DTC1	DTC1_DTCEND	DTC1 transfer end
0x070	PVD	PVD_PVD1	Voltage monitor 1 interrupt
0x071		PVD_PVD2	Voltage monitor 2 interrupt
0x076	MOSC	MOSC_STOP	Mail Clock oscillation stop
0x080	ULPT0	ULPT0_ULPTI	Underflow
0x081		ULPT0_ULPTCMAI	Compare match A
0x082		ULPT0_ULPTCMBI	Compare match B
0x083	ULPT1	ULPT1_ULPTI	Underflow
0x084		ULPT1_ULPTCMAI	Compare match A
0x085		ULPT1_ULPTCMBI	Compare match B
0x086	AGT0	AGT0_AGTI	AGT interrupt
0x087		AGT0_AGTCMAI	Compare match A
0x088		AGT0_AGTCMBI	Compare match B
0x089	AGT1	AGT1_AGTI	AGT interrupt
0x08A		AGT1_AGTCMAI	Compare match A
0x08B		AGT1_AGTCMBI	Compare match B
0x092	IWDT	IWDT_NMIUNDF	IWDT underflow
0x093	WDT	WDT0_NMIUNDF	WDT underflow
0x094		WDT1_NMIUNDF	WDT underflow
0x096	RTC	RTC_PRD	Periodic interrupt
0x09C	IIC0	IIC0_RXI	Receive data full
0x09D		IIC0_TXI	Transmit data empty
0x09E		IIC0_TEI	Transmit end
0x09F		IIC0_EEI	Transfer error

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (3 of 9)

Event number	Interrupt request source	Name	Description
0x0A1	IIC1	IIC1_RXI	Receive data full
0x0A2		IIC1_TXI	Transmit data empty
0x0A3		IIC1_TEI	Transmit end
0x0A4		IIC1_EEI	Transfer error
0x0A6	IIC2	IIC2_RXI	Receive data full
0x0A7		IIC2_TXI	Transmit data empty
0x0A8		IIC2_TEI	Transmit end
0x0A9		IIC2_EEI	Transfer error
0x0C6	ACMPHS	ACMP_HS0*1	Comparator interrupt 0
0x0C7		ACMP_HS1*1	Comparator interrupt 1
0x0C8		ACMP_HS2*1	Comparator interrupt 2
0x0C9		ACMP_HS3*1	Comparator interrupt 3
0x0CC	ELC	ELC_SWEVT0	Software event 0
0x0CD		ELC_SWEVT1	Software event 1
0x0CE		ELC_SWEVT2	Software event 2
0x0CF		ELC_SWEVT3	Software event 3
0x0D0	I/O Port	IOPORT_GROUP1	Port 1 event
0x0D1		IOPORT_GROUP2	Port 2 event
0x0D2		IOPORT_GROUP3	Port 3 event
0x0D3		IOPORT_GROUP4	Port 4 event
0x180	GPT	GPT_UVWEDGE	UVW edge event
0x181	GPT0	GPT0_CCMPA	Compare match A
0x182		GPT0_CCMPB	Compare match B
0x183		GPT0_CMPC	Compare match C
0x184		GPT0_CMPD	Compare match D
0x185		GPT0_CMPE	Compare match E
0x186		GPT0_CMPF	Compare match F
0x187		GPT0_OVF	Overflow
0x188		GPT0_UDF	Underflow
0x189		GPT0_PC	Cycle count function end
0x18A	GPT1	GPT1_CCMPA	Compare match A
0x18B		GPT1_CCMPB	Compare match B
0x18C		GPT1_CMPC	Compare match C
0x18D		GPT1_CMPD	Compare match D
0x18E		GPT1_CMPE	Compare match E
0x18F		GPT1_CMPF	Compare match F
0x190		GPT1_OVF	Overflow
0x191		GPT1_UDF	Underflow
0x192		GPT1_PC	Cycle count function end

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (4 of 9)

Event number	Interrupt request source	Name	Description
0x193	GPT2	GPT2_CCMPA	Compare match A
0x194		GPT2_CCMPB	Compare match B
0x195		GPT2_CMPC	Compare match C
0x196		GPT2_CMPD	Compare match D
0x197		GPT2_CMPE	Compare match E
0x198		GPT2_CMPF	Compare match F
0x199		GPT2_OVF	Overflow
0x19A		GPT2_UDF	Underflow
0x19B		GPT2_PC	Cycle count function end
0x19C	GPT3	GPT3_CCMPA	Compare match A
0x19D		GPT3_CCMPB	Compare match B
0x19E		GPT3_CMPC	Compare match C
0x19F		GPT3_CMPD	Compare match D
0x1A0		GPT3_CMPE	Compare match E
0x1A1		GPT3_CMPF	Compare match F
0x1A2		GPT3_OVF	Overflow
0x1A3		GPT3_UDF	Underflow
0x1A4		GPT3_PC	Cycle count function end
0x1A5	GPT4	GPT4_CCMPA	Compare match A
0x1A6		GPT4_CCMPB	Compare match B
0x1A7		GPT4_CMPC	Compare match C
0x1A8		GPT4_CMPD	Compare match D
0x1A9		GPT4_CMPE	Compare match E
0x1AA		GPT4_CMPF	Compare match F
0x1AB		GPT4_OVF	Overflow
0x1AC		GPT4_UDF	Underflow
0x1AE	GPT5	GPT5_CCMPA	Compare match A
0x1AF		GPT5_CCMPB	Compare match B
0x1B0		GPT5_CMPC	Compare match C
0x1B1		GPT5_CMPD	Compare match D
0x1B2		GPT5_CMPE	Compare match E
0x1B3		GPT5_CMPF	Compare match F
0x1B4		GPT5_OVF	Overflow
0x1B5		GPT5_UDF	Underflow
0x1B7	GPT6	GPT6_CCMPA	Compare match A
0x1B8		GPT6_CCMPB	Compare match B
0x1B9		GPT6_CMPC	Compare match C
0x1BA		GPT6_CMPD	Compare match D
0x1BB		GPT6_CMPE	Compare match E
0x1BC		GPT6_CMPF	Compare match F
0x1BD		GPT6_OVF	Overflow
0x1BE		GPT6_UDF	Underflow

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (5 of 9)

Event number	Interrupt request source	Name	Description
0x1C0	GPT7	GPT7_CCMPA	Compare match A
0x1C1		GPT7_CCMPB	Compare match B
0x1C2		GPT7_CMPC	Compare match C
0x1C3		GPT7_CMPD	Compare match D
0x1C4		GPT7_CMPE	Compare match E
0x1C5		GPT7_CMPF	Compare match F
0x1C6		GPT7_OVF	Overflow
0x1C7		GPT7_UDF	Underflow
0x1C9	GPT8	GPT8_CCMPA	Compare match A
0x1CA		GPT8_CCMPB	Compare match B
0x1CB		GPT8_CMPC	Compare match C
0x1CC		GPT8_CMPD	Compare match D
0x1CD		GPT8_CMPE	Compare match E
0x1CE		GPT8_CMPF	Compare match F
0x1CF		GPT8_OVF	Overflow
0x1D0		GPT8_UDF	Underflow
0x1D2	GPT9	GPT9_CCMPA	Compare match A
0x1D3		GPT9_CCMPB	Compare match B
0x1D4		GPT9_CMPC	Compare match C
0x1D5		GPT9_CMPD	Compare match D
0x1D6		GPT9_CMPE	Compare match E
0x1D7		GPT9_CMPF	Compare match F
0x1D8		GPT9_OVF	Overflow
0x1D9		GPT9_UDF	Underflow
0x1DB	GPT10	GPT10_CCMPA	Compare match A
0x1DC		GPT10_CCMPB	Compare match B
0x1DD		GPT10_CMPC	Compare match C
0x1DE		GPT10_CMPD	Compare match D
0x1DF		GPT10_CMPE	Compare match E
0x1E0		GPT10_CMPF	Compare match F
0x1E1		GPT10_OVF	Overflow
0x1E2		GPT10_UDF	Underflow
0x1E3		GPT10_PC	Cycle count function end
0x1E4	GPT11	GPT11_CCMPA	Compare match A
0x1E5		GPT11_CCMPB	Compare match B
0x1E6		GPT11_CMPC	Compare match C
0x1E7		GPT11_CMPD	Compare match D
0x1E8		GPT11_CMPE	Compare match E
0x1E9		GPT11_CMPF	Compare match F
0x1EA		GPT11_OVF	Overflow
0x1EB		GPT11_UDF	Underflow
0x1EC		GPT11_PC	Cycle count function end

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (6 of 9)

Event number	Interrupt request source	Name	Description
0x1ED	GPT12	GPT12_CCMPA	Compare match A
0x1EE		GPT12_CCMPB	Compare match B
0x1EF		GPT12_CMPC	Compare match C
0x1F0		GPT12_CMPD	Compare match D
0x1F1		GPT12_CMPE	Compare match E
0x1F2		GPT12_CMPF	Compare match F
0x1F3		GPT12_OVF	Overflow
0x1F4		GPT12_UDF	Underflow
0x1F5		GPT12_PC	Cycle count function end
0x1F6	GPT13	GPT13_CCMPA	Compare match A
0x1F7		GPT13_CCMPB	Compare match B
0x1F8		GPT13_CMPC	Compare match C
0x1F9		GPT13_CMPD	Compare match D
0x1FA		GPT13_CMPE	Compare match E
0x1FB		GPT13_CMPF	Compare match F
0x1FC		GPT13_OVF	Overflow
0x1FD		GPT13_UDF	Underflow
0x1FE		GPT13_PC	Cycle count function end
0x2B4	ESWM	GPTP_PTPOUT0	PTP Pulse Out 0
0x2B5		GPTP_PTPOUT1	PTP Pulse Out 1
0x2B6		GPTP_PTPOUT2	PTP Pulse Out 2
0x2B7		GPTP_PTPOUT3	PTP Pulse Out 3
0x2B8		GPTP_MATCH0	Media Clock Recovery Match 0
0x2B9		GPTP_MATCH1	Media Clock Recovery Match 1
0x2BA	ESC	ETHCSI0	Sync0 Interrupt
0x2BB		ETHCSI1	Sync1 Interrupt
0x2C4	SCI0	SCI0_RXI*2	Receive data full
0x2C5		SCI0_TXI*2	Transmit data empty
0x2C6		SCI0_TEI*2	Transmit end
0x2C7		SCI0_ERI	Receive error
0x2C8		SCI0_AED	Effective edge detection
0x2CA		SCI0_AM	Address match event
0x2CB	SCI1	SCI1_RXI*2	Receive data full
0x2CC		SCI1_TXI*2	Transmit data empty
0x2CD		SCI1_TEI*2	Transmit end
0x2CE		SCI1_ERI	Receive error
0x2CF		SCI1_AED	Effective edge detection
0x2D1		SCI1_AM	Address match event

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (7 of 9)

Event number	Interrupt request source	Name	Description
0x2D2	SCI2	SCI2_RXI* ²	Receive data full
0x2D3		SCI2_TXI* ²	Transmit data empty
0x2D4		SCI2_TEI* ²	Transmit end
0x2D5		SCI2_ERI	Receive error
0x2D6		SCI2_AED	Effective edge detection
0x2D8		SCI2_AM	Address match event
0x2D9	SCI3	SCI3_RXI* ²	Receive data full
0x2DA		SCI3_TXI* ²	Transmit data empty
0x2DB		SCI3_TEI* ²	Transmit end
0x2DC		SCI3_ERI	Receive error
0x2DD		SCI3_AED	Effective edge detection
0x2DF		SCI3_AM	Address match event
0x2E0	SCI4	SCI4_RXI* ²	Receive data full
0x2E1		SCI4_TXI* ²	Transmit data empty
0x2E2		SCI4_TEI* ²	Transmit end
0x2E3		SCI4_ERI	Receive error
0x2E4		SCI4_AED	Effective edge detection
0x2E6		SCI4_AM	Address match event
0x2E7	SCI5	SCI5_RXI* ²	Receive data full
0x2E8		SCI5_TXI* ²	Transmit data empty
0x2E9		SCI5_TEI* ²	Transmit end
0x2EA		SCI5_ERI	Receive error
0x2EB		SCI5_AED	Effective edge detection
0x2ED		SCI5_AM	Address match event
0x2EE	SCI6	SCI6_RXI* ²	Receive data full
0x2EF		SCI6_TXI* ²	Transmit data empty
0x2F0		SCI6_TEI* ²	Transmit end
0x2F1		SCI6_ERI	Receive error
0x2F2		SCI6_AED	Effective edge detection
0x2F4		SCI6_AM	Address match event
0x2F5	SCI7	SCI7_RXI* ²	Receive data full
0x2F6		SCI7_TXI* ²	Transmit data empty
0x2F7		SCI7_TEI* ²	Transmit end
0x2F8		SCI7_ERI	Receive error
0x2F9		SCI7_AED	Effective edge detection
0x2FB		SCI7_AM	Address match event

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (8 of 9)

Event number	Interrupt request source	Name	Description
0x2FC	SCI8	SCI8_RXI*2	Receive data full
0x2FD		SCI8_TXI*2	Transmit data empty
0x2FE		SCI8_TEI*2	Transmit end
0x2FF		SCI8_ERI	Receive error
0x300		SCI8_AED	Effective edge detection
0x302		SCI8_AM	Address match event
0x303	SCI9	SCI9_RXI*2	Receive data full
0x304		SCI9_TXI*2	Transmit data empty
0x305		SCI9_TEI*2	Transmit end
0x306		SCI9_ERI	Receive error
0x307		SCI9_AED	Effective edge detection
0x309		SCI9_AM	Address match event
0x318	SPI0	SPI0_SPRI	Receive buffer full
0x319		SPI0_SPTI	Transmit buffer empty
0x31A		SPI0_SPII	Idle
0x31B		SPI0_SPEI	Error
0x31C		SPI0_SPCEND	Communication complete event
0x31D	SPI1	SPI1_SPRI	Receive buffer full
0x31E		SPI1_SPTI	Transmit buffer empty
0x31F		SPI1_SPII	Idle
0x320		SPI1_SPEI	Error
0x321		SPI1_SPCEND	Transmission complete event
0x33A	I3C	I3C_RESP	Normal Response queue full
0x33B		I3C_CMD	Normal Command queue empty
0x33C		I3C_IBI	Normal IBI Queue Empty/Full
0x33D		I3C_RX	Normal Rx Data buffer full
0x33E		I3C_TX	Normal Tx Data buffer empty
0x33F		I3C_RCV	Normal Receive Status queue full
0x340		I3C_HRESP	High priority response queue full
0x341		I3C_HCMD	High priority command queue empty
0x342		I3C_HRX	High priority Rx Data buffer full
0x343		I3C_HTX	High priority Tx Data buffer empty
0x344		I3C_TEND	Transmit end
0x345		I3C_EEI	Transfer error or event occurrence
0x346		I3C_STEV	Synchronous Timing
0x347		I3C_MREFOVF	MREF Counter Overflow
0x348		I3C_MREFCPT	MREF Capture
0x349		I3C_AMEV	Additional Master-initiated bus Event

Table 19.3 Association between event signal names set in ELSRn.ELS[9:0] bits and signal numbers (9 of 9)

Event number	Interrupt request source	Name	Description
0x34D	ADC16H	ADC_ADI0	A/D scan end interrupt for group 0
0x34E		ADC_ADI1	A/D scan end interrupt for group 1
0x34F		ADC_ADI2	A/D scan end interrupt for group 2
0x350		ADC_ADI3	A/D scan end interrupt for group 3
0x351		ADC_ADI4	A/D scan end interrupt for group 4
0x352		ADC_ADI5	A/D scan end interrupt for group 5
0x353		ADC_ADI6	A/D scan end interrupt for group 6
0x354		ADC_ADI7	A/D scan end interrupt for group 7
0x355		ADC_ADI8	A/D scan end interrupt for group 8
0x361		ADC_CCMPM0	Composite condition compare 0 match
0x362		ADC_CCMPUM0	Composite condition compare 0 mismatch
0x369		ADC_CCMPM1	Composite condition compare 1 match
0x36A		ADC_CCMPUM1	Composite condition compare 1 mismatch
0x36F	DOC	DOC_DOPCI	Data operation circuit interrupt

Note 1. Only pulse (edge detection) is supported.

Note 2. This event is not supported in FIFO mode.

19.2.4 ELCSARA : Event Link Controller Security Attribution Register A

Base ELC = 0x4020_1000
address: ELC_NS = 0x5020_1000

Offset
address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	ELSE GR3	ELSE GR2	ELSE GR1	ELSE GR0	ELCR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ELCR	Event Link Controller Register Security Attribution Target register: ELCR 0: Secure 1: Non-secure	R/W
1	ELSEGR0	Event Link Software Event Generation Register 0 Security Attribution 0: Secure 1: Non-secure	R/W
2	ELSEGR1	Event Link Software Event Generation Register 1 Security Attribution 0: Secure 1: Non-secure	R/W
3	ELSEGR2	Event Link Software Event Generation Register 2 Security Attribution 0: Secure 1: Non-secure	R/W
4	ELSEGR3	Event Link Software Event Generation Register 3 Security Attribution 0: Secure 1: Non-secure	R/W

Bit	Symbol	Function	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

Note: This register is write-protected by PRCR_S register.

This register specifies the security attribution for the ELCR, ELSEGR0, ELSEGR1, ELSEGR2, and ELSEGR3 registers.

19.2.5 ELCSARB : Event Link Controller Security Attribution Register B

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ELSR[31:30]	—	—	ELSR[27:19]										—	ELSR[17:16]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ELSR[15:12]				—	—	—	—	ELSR[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	ELSR7 to ELSR0	Event Link Setting Register n Security Attribution Target register: ELSRn (n = 0 to 7) 0: Secure 1: Non-secure	R/W
11:8	—	These bits are read as 0. The write value should be 0.	R/W
17:12	ELSR17 to ELSR12	Event Link Setting Register n Security Attribution Target register: ELSRn (n = 12 to 17) 0: Secure 1: Non-secure	R/W
18	—	This bit is read as 0. The write value should be 0.	R/W
27:19	ELSR27 to ELSR19	Event Link Setting Register n Security Attribution Target register: ELSRn (n = 19 to 27) 0: Secure 1: Non-secure	R/W
29:28	—	These bits are read as 0. The write value should be 0.	R/W
31:30	ELSR31 to ELSR30	Event Link Setting Register n Security Attribution Target register: ELSRn (n = 30, 31) 0: Secure 1: Non-secure	R/W

Note: S-TYPE-1, P-TYPE-1

Note: This register is write-protected by PRCR_S register.

This register specifies the security attribution for the Register ELSRn (n = 0 to 7, 12 to 17, 19 to 27, 30, 31).

19.2.6 ELCSARC : Event Link Controller Security Attribution Register C

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x108

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	ELSR[52:48]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ELSR[47:32]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
20:0	ELSR[52:32]	Event Link Setting Register n Security Attribution Target register: ELSRn (n = 32 to 52) 0: Secure 1: Non-secure	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1, P-TYPE-1

Note: This register is write-protected by PRCR_S register.

This register specifies the security attribution for the Register ELSRn (n = 32 to 52).

19.2.7 ELCPARA : Event Link Controller Privilege Attribution Register A

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x110

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	ELSE GR3	ELSE GR2	ELSE GR1	ELSE GR0	ELCR
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
0	ELCR	Event Link Controller Register Privilege Attribution Target register: ELCR 0: Privileged 1: Unprivileged	R/W
1	ELSEGR0	Event Link Software Event Generation Register 0 Privilege Attribution Target register: ELSEGR0 0: Privileged 1: Unprivileged	R/W
2	ELSEGR1	Event Link Software Event Generation Register 1 Privilege Attribution Target register: ELSEGR1 0: Privileged 1: Unprivileged	R/W

Bit	Symbol	Function	R/W
3	ELSEGR2	Event Link Software Event Generation Register 2 Privilege Attribution Target register: ELSEGR2 0: Privileged 1: Unprivileged	R/W
4	ELSEGR3	Event Link Software Event Generation Register 3 Privilege Attribution Target register: ELSEGR3 0: Privileged 1: Unprivileged	R/W
31:5	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-2, P-TYPE-1

Note: This register is write-protected by PRCR_S and PRCR_NS register.

This register specifies the privilege attribution for the Register ELCR, ELSEGR0, ELSEGR1, ELSEGR2, and ELSEGR3 registers.

19.2.8 ELCPARB : Event Link Controller Privilege Attribution Register B

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x114

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	ELSR[31:30]	—	—	ELSR[27:19]										—	ELSR[17:16]	
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ELSR[15:12]				—	—	—	—	ELSR[7:0]							
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
7:0	ELSR7 to ELSR0	Event Link Setting Register n Privilege Attribution Target register: ELSRn (n = 0 to 7) 0: Privileged 1: Unprivileged	R/W
11:8	—	These bits are read as 1. The write value should be 1.	R/W
17:12	ELSR17 to ELSR12	Event Link Setting Register n Privilege Attribution Target register: ELSRn (n = 12 to 17) 0: Privileged 1: Unprivileged	R/W
18	—	This bit is read as 1. The write value should be 1.	R/W
27:19	ELSR27 to ELSR19	Event Link Setting Register n Privilege Attribution Target register: ELSRn (n = 19 to 27) 0: Privileged 1: Unprivileged	R/W
29:28	—	These bits are read as 1. The write value should be 1.	R/W
31:30	ELSR31 to ELSR30	Event Link Setting Register n Privilege Attribution Target register: ELSRn (n = 30, 31) 0: Privileged 1: Unprivileged	R/W

Note: S-TYPE-2, P-TYPE-1

Note: This register is write-protected by PRCR_S and PRCR_NS register.

This register specifies the privilege attribution for the Register ELSRn (n = 0 to 7, 12 to 17, 19 to 27, 30, 31) registers.

19.2.9 ELCPARC : Event Link Controller Privilege Attribution Register C

Base address: ELC = 0x4020_1000
ELC_NS = 0x5020_1000

Offset address: 0x118

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	ELSR[52:48]				
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ELSR[47:32]															
Value after reset:	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Bit	Symbol	Function	R/W
20:0	ELSR[52:32]	Event Link Setting Register n Privilege Attribution Target register: ELSRn (n = 32 to 52) 0: Privileged 1: Unprivileged	R/W
31:21	—	These bits are read as 1. The write value should be 1.	R/W

Note: S-TYPE-2, P-TYPE-1

Note: This register is write-protected by PRCR_S and PRCR_NS register.

This register specifies the privilege attribution for the Register ELSRn (n = 32 to 52) registers.

19.3 Operation

19.3.1 Relation Between Interrupt Handling and Event Linking

Event number for an event link is the same as that for the associated interrupt source. For information on generating event signals, see the explanation in the chapter for each event source module.

19.3.2 Linking Events

When an event occurs and that event is already set as a trigger in the Event Link Setting Register (ELSRn), the associated module is activated. The operation of the module must be set up in advance. [Table 19.4](#) lists the operations of modules when an event occurs.

Table 19.4 Module operations when event occurs (1 of 2)

Module	Operations when event is input
GPT	- Start counting - Stop counting - Clear counting - Up counting - Down counting - Input capture
DAC12	Start D/A conversion
I/O Ports	- Change pin output based on the EORR (reset) or EOSR (set) - Latch pin state to EIDR - The following ports can be used for the ELC: PORT 1 PORT 2 PORT 3 PORT 4
I3C	Start operation
ADC16H	Start A/D conversion

Table 19.4 Module operations when event occurs (2 of 2)

Module	Operations when event is input
DTC	Start DTC data transfer
ESWM	Media clock capture event occurs
ESC	Capture current value of EtherCAT Slave Controller DC
DSMIF	- Current data capture trigger 0 to 5 - Decimation dividing counter initialization trigger 0 to 2

19.3.3 Example of Procedure for Linking Events

To link events:

1. Set the operation of the module for which an event is to be linked.
2. Set the appropriate ELSRn.ELS[9:0] bits for the module to be linked.
3. Set the ELCR.ELCON bit to 1 to enable linkage of all events.
4. Configure the module from which an event is output and activate the module. The link between the two modules is now active.
5. To stop event linkage of modules individually, set 0 to the ELSRn.ELS[9:0] bits associated with the modules. To stop linkage of all the events, set the ELCR.ELCON bit to 0.

If event link output from the PVD is to be used, set the ELC after setting the PVD. To disable the PVD, do so after setting 0x00 to the associated ELSRn register.

19.4 Usage Notes

19.4.1 Linking DMAC/DTC Transfer End Signals as Events

When linking the DMAC/DTC transfer end signals as events, do not set the same peripheral module as the DMAC/DTC transfer destination and event link destination. If set, the peripheral module might be started before DMAC/DTC transfer to the peripheral module is complete.

19.4.2 Setting Clocks

To link events, you must enable the ELC and the related modules. The modules cannot operate if the related modules are in the module-stop state or in low power mode in which the module is stopped (CPU Deep Sleep mode, Software Standby mode or Deep Software Standby mode). For more information, see [Table 19.3](#) and [section 11, Low Power Mode](#).

19.4.3 Module-Stop Function Setting

The Module Stop Control Register C (MSTPCRC) can enable or disable ELC operation. The ELC is initially stopped after reset. Releasing the module-stop state enables access to the registers. The ELCON bit must be set to 0 before disabling ELC operation using the Module Stop Control Register. For more information, see [Table 19.3](#) and [section 11, Low Power Mode](#).

19.4.4 ELC Delay Time

In [Figure 19.2](#), module A accesses module B through the ELC. There is a delay time in the ELC between module A and module B. [Table 19.5](#) shows the ELC delay time.

[Table 19.6](#) shows the clock domain of each module A. [Table 19.7](#) shows the clock domain of each module B.

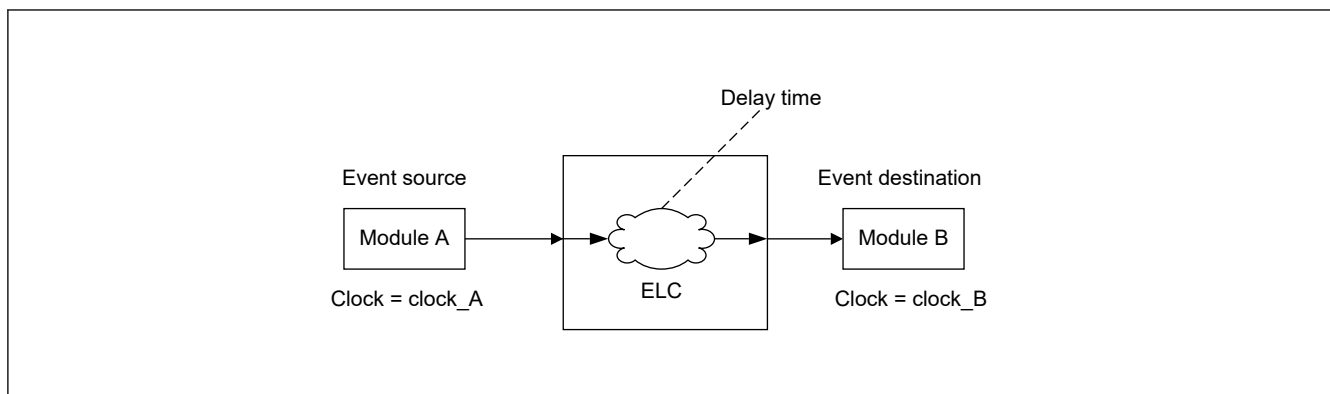


Figure 19.2 ELC delay time

Table 19.5 ELC delay time

Clock domain	Clock frequency	ELC delay time
clock_A = clock_B	clock_A = clock_B	0 cycle
clock_A ≠ clock_B	clock_A = clock_B	1 cycle to 2 cycles
	clock_A > clock_B	1 cycle to 2 cycles of clock_B
	clock_A < clock_B	1 cycle to 2 cycles of clock_A
clock_A ≠ clock_B	clock_A = clock_B ²	0 cycle
(Asynchronous) ^{*1}	—	5 cycles of clock_A + 4 cycles of clock_B

Note 1. When GPTCLK is selected as the GPT count clock, use this formula.

When ADC16H is destination and ADCCLK or GPTCLK is selected as the A/D conversion clock, use this formula.

When DSMIF is destination and DSMIFCLK or GPTCLK is selected as the DSMIF operation clock, use this formula.

Note 2. In case GPTCLK is selected as the GPT count clock, ADC conversion clock and DSMIF operation clock.

Table 19.6 Clock domain of each module A

Module A	Clock domain of clock_A
SCI, SPI, I3C, ADC16H, DOC, ESWM, ESC	PCLKA
Port, DMAC, PVD1, PVD2, MOSC, ULPT, AGT, IWDT, WDT, IIC, ACMPHS, ELC, I/O Ports	PCLKB
GPT	GPTCLK or PCLKD

Table 19.7 Clock domain of each module B

Module B	Clock domain of clock_B
I3C, ESWM, ESC	PCLKA
I/O Ports, DAC12	PCLKB
GPT	GPTCLK or PCLKD
ADC16H	ADCLK, GPTCLK, or PCLKA
DSMIF	DSMIFCLK, GPTCLK, or PCLKA